

CONNECTED STRATEGY

**BUILDING CONTINUOUS CUSTOMER
RELATIONSHIPS FOR
COMPETITIVE ADVANTAGE**

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TERWIESCH**

Contents

FIGURE 1-1	4
FIGURE 2-1	4
FIGURE 2-2	5
FIGURE 2-3	5
TABLE 2-1	6
TABLE 2-2	6
WORKSHEET 3-1	7
WORKSHEET 3-2	8
FIGURE 4-1	9
FIGURE 4-2	10
FIGURE 4-3	11
FIGURE 4-4	12
FIGURE 4-5	13
TABLE 4-1	14
TABLE 4-2	15
FIGURE 5-1	16
FIGURE 5-2	16
FIGURE 5-3	16
FIGURE 5-4	17
TABLE 5-1	17
FIGURE 5-5	18
WORKSHEET 6-1	19
WORKSHEET 6-2	20
WORKSHEET 6-3	21
WORKSHEET 6-4	22
WORKSHEET 6-5	23
WORKSHEET 6-6	24
FIGURE 7-1	25
FIGURE 7-2	25
FIGURE 7-3	26
FIGURE 7-4	26
FIGURE 7-5	27
FIGURE 7-6	28
TABLE 9-1	29
FIGURE 9-1	31
TABLE 9-2	32
WORKSHEET 10-1	33
WORKSHEET 10-2	34

WORKSHEET 10-3	35
WORKSHEET 10-4	36
WORKSHEET 10-5	37

FIGURE 1-1

The connected strategy

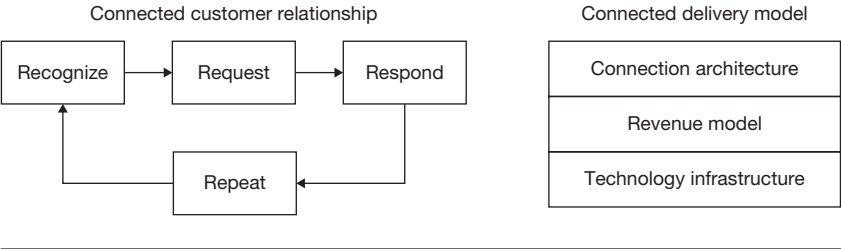


FIGURE 2-1

Efficiency frontier

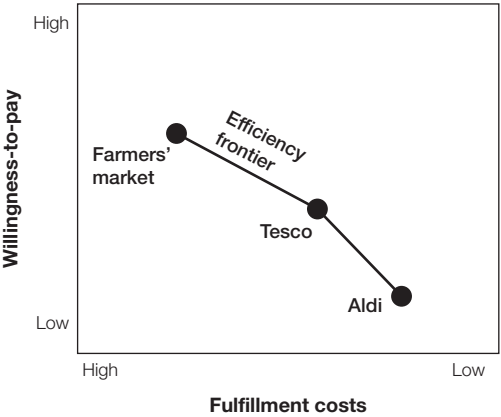


FIGURE 2-2

Pushing out the efficiency frontier

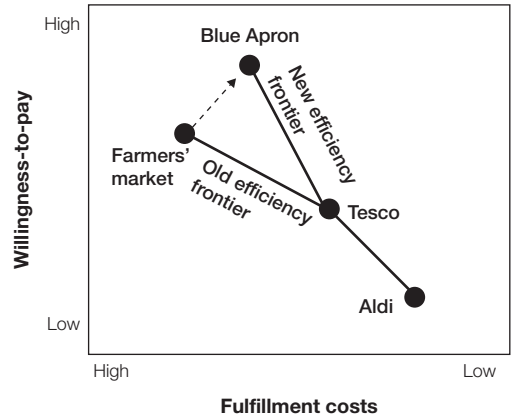


FIGURE 2-3

New efficiency frontier in the grocery industry

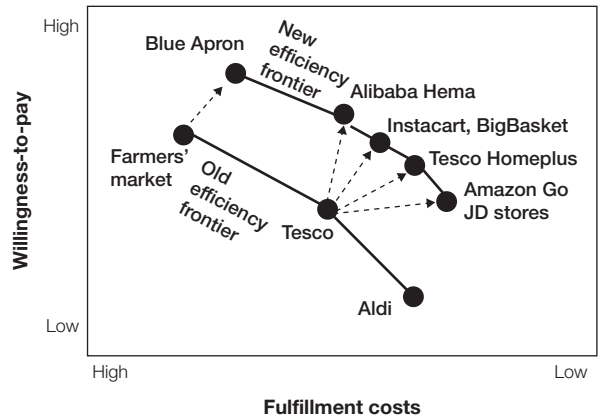


TABLE 2-1

Benefits of creating deeper connections

Connection	Status quo cab	“Connected cab”
Passenger to vehicle	Flagging, dispatcher, taxi stand	App
Payment	Cash, credit card	Automated in-app purchase
Vehicle to vehicle	Dispatcher, two-way radio	GPS tracking, fleet management system
Improvements		• More convenient ordering • No wasted time for payment • Better routing

TABLE 2-2

Benefits of creating new connections

Connection	Status quo	Ride hailing
Passenger to vehicle	Flagging, dispatcher, taxi stand	App
Payment	Cash, credit card	Automated in-app purchase
Vehicle to vehicle	Dispatcher, two-way radio	GPS tracking, fleet management system
Vehicle to vehicle outside the fleet	—	Connects all vehicles on the platform
Idle drivers and unused vehicles	—	Connects to drivers who are presently not driving
Passenger to passenger	—	Crowdsource driver reputations
Driver to driver	—	Passenger ratings leading to driver safety
Improvements	—	• Higher vehicle utilization • No medallion needed • No wasted time for driver to pick up and drop off car at the beginning and end of shift

Identify willingness-to-pay drivers of your customers

Consumption utility: How happy is the customer with the product or service?	Accessibility: How easy is it for the customer to get the product or service?	Cost of ownership: How much does it cost the customer to use and maintain the product?
Performance	Location	Usage costs over product life
Fit	Timing	Maintenance costs over product life

The efficiency frontier for your industry

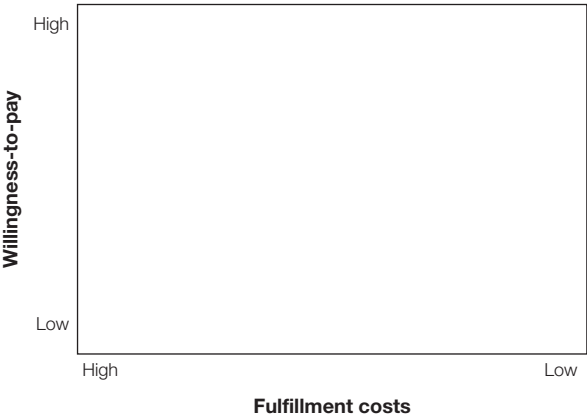


FIGURE 4-1

The three phases of the customer journey

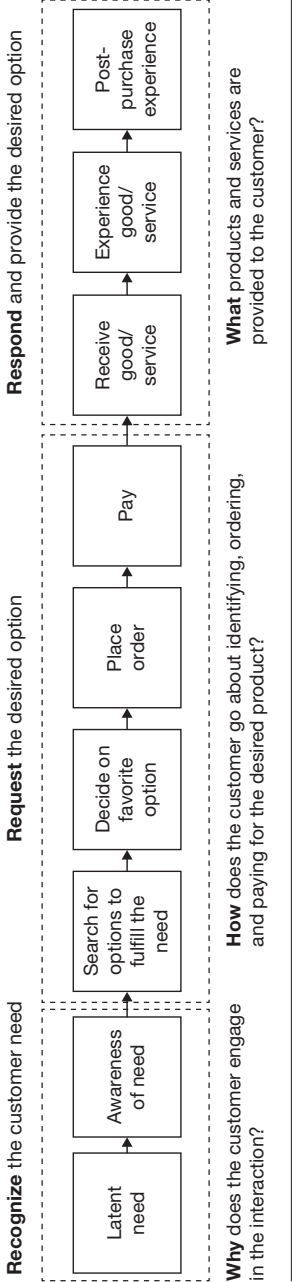


FIGURE 4-2

The respond-to-desire connected customer experience

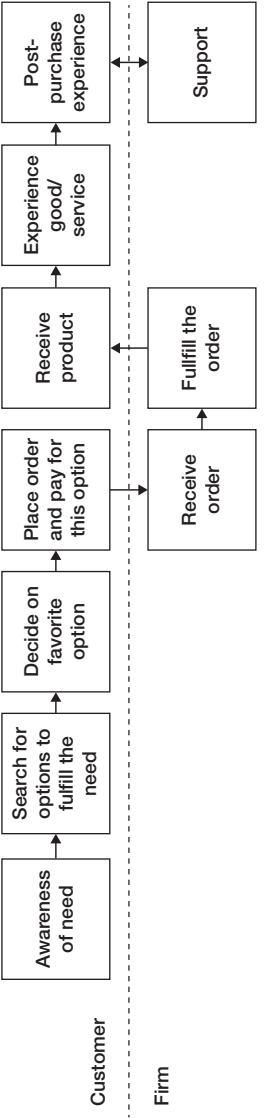


FIGURE 4-3

The curated offering connected customer experience

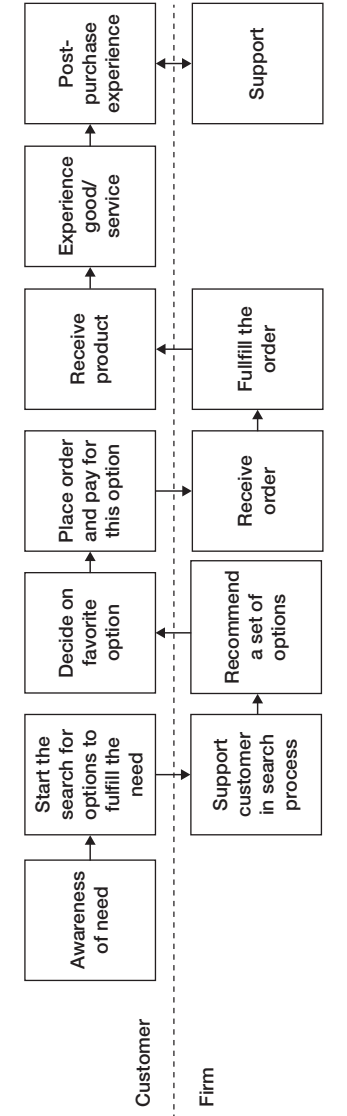


FIGURE 4-4

The coach behavior connected customer experience

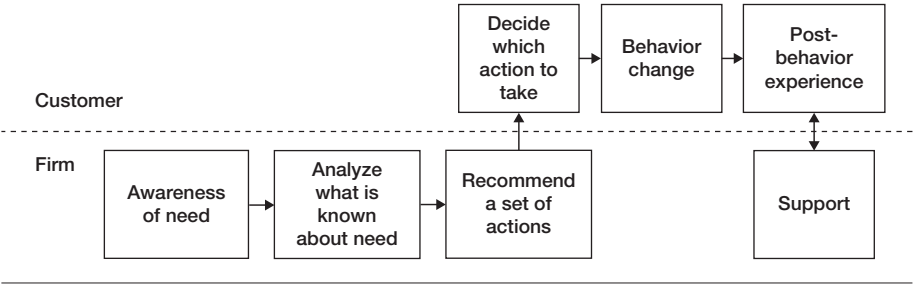


FIGURE 4-5

The automatic execution connected customer experience

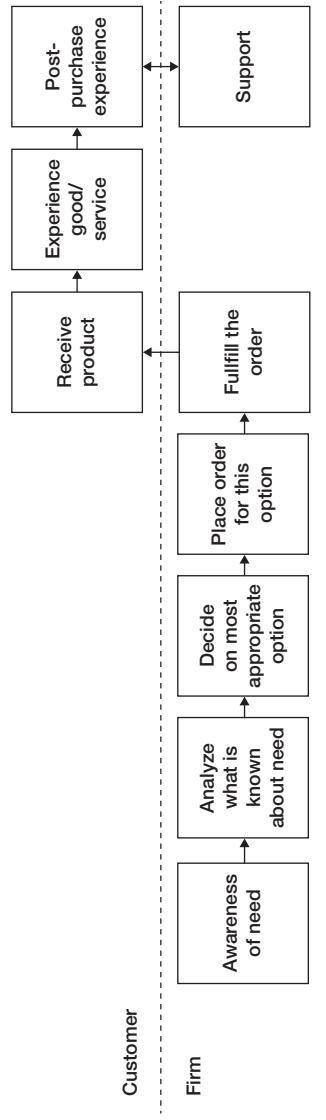


TABLE 4-1

Information flow dimensions for different customer experiences

	Buy-what-we-have	Respond-to-desire	Curated offering	Coach behavior	Automatic execution
Trigger of information flow	Customer	Customer	Customer	Firm	Firm
Frequency of information flow	Episodic	Episodic	Increased	Continuous	Continuous
Richness of information flow	Low	Low	Medium	High	High
Customer effort to send information	High	Reduced	Reduced	None	None
Information processing to find preferences	All preferences expressed explicitly by the customer	All preferences expressed explicitly by the customer	Firm makes recommendations to the customer	Firm attempts to drive very specific customer action	Firm autonomously takes action for the customer

TABLE 4-2

Domains of application for the four connected customer experiences

Type of connected customer experience	Description	Key capability	Works best when	Works best for
Respond-to-desire	Customer expresses what she wants and when	Fast and efficient response to the incoming orders	Customers are knowledgeable	Customers who don't want to share too much data and want to be in control
Curated offering	Firm offers a tailored menu of options to the customer; final choice rests with the customer	Making good recommendations	The set of options is large, potentially overwhelming for the customer	Customers who are fine sharing some data but still want to have final say in decisions
Coach behavior	Firm nudges customer to adjust her immediate preferences, either to obtain some larger goal or to reduce fulfillment costs	Understanding latent customer needs and how they are (not) related to the immediate actions	Customers have inertia and other biases keeping them from achieving what is best for them	Customers who do not mind sharing personal data and an environment that influences their behavior
Automatic execution	Firm monitors, then executes the fulfillment process without action from customer	Monitoring the customer and translating incoming data into action	Customer behavior is very predictable and costs of mistakes are small	Customers who do not mind sharing personal data and having firms making decisions for them

FIGURE 5-1

The traditional efficiency frontier in education

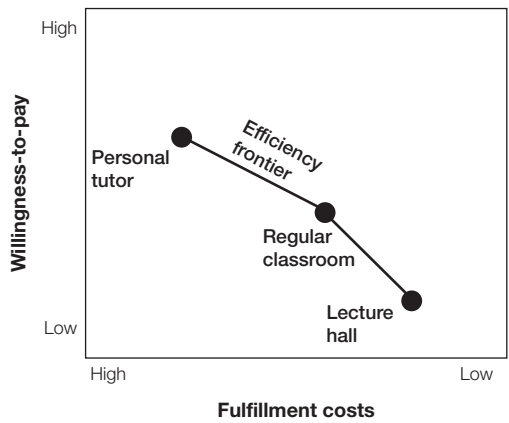


FIGURE 5-2

Learning at the level of the individual customer

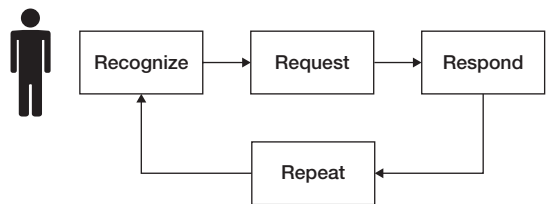


FIGURE 5-3

Population-level learning

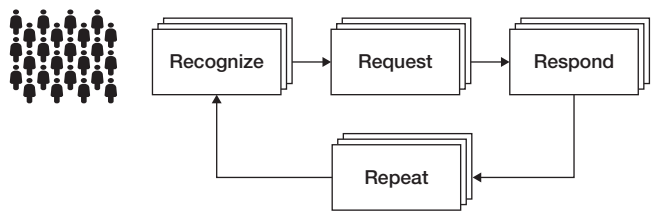
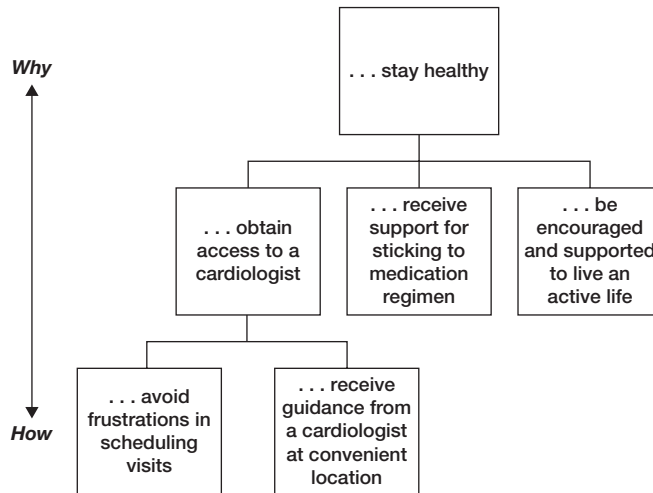


FIGURE 5-4

The why-how ladder for cardiology problems

In the eyes of the customer, the purpose of the relationship with our firm is to . . .

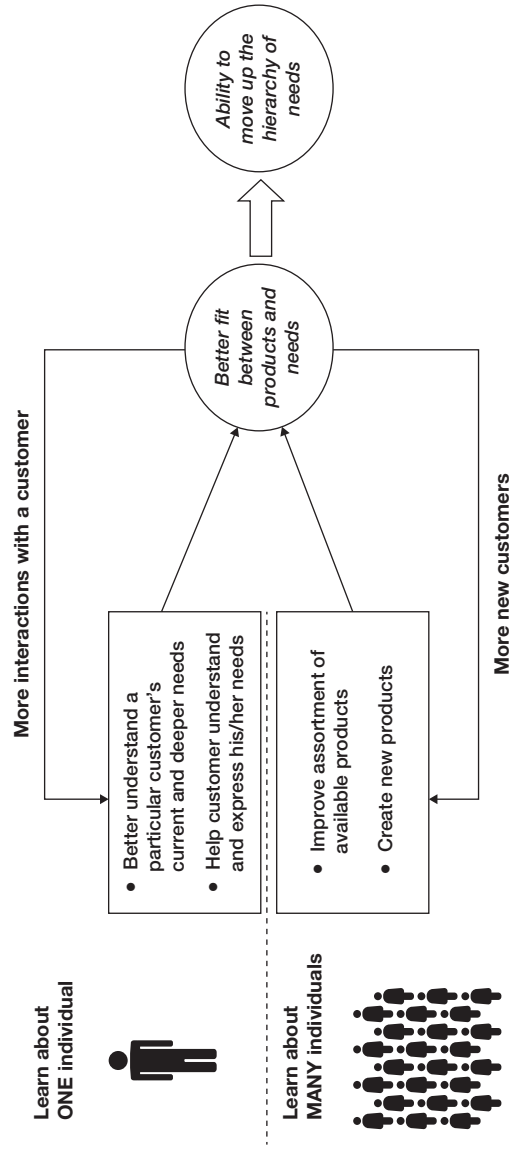
**TABLE 5-1**

The four levels of customization created by the repeat dimension

Level	Impact on willingness-to-pay	Impact on cost
Level 1: Create unified customer experiences across episodes	Customer is treated as one person across channels and transactions	Avoids manual weaving together of experiences
Level 2: Improve customization based on past interactions	Ability to identify offerings that address the willingness-to-pay drivers most important to the particular customer	Avoids costly iterations in case of failure to fulfill the need
Level 3: Learn at the population level to enhance product offerings	Higher-valued offerings based on inferring customer needs	Data-driven approach to innovation
Level 4: Become a trusted partner to the customer	Addressing more fundamental needs allows for alternative solutions and early interventions	More efficient use of resources, as solution space is broadened

FIGURE 5-5

The positive learning feedback loops created by the repeat dimension



Customer journey

Why does the customer engage in the interaction?	How does the customer go about identifying, ordering, and paying for the desired product?	What products and services are provided to the customer?
Latent need	Awareness of need	Search for options
	Decide on options	Order and pay
	Receive	Experience good/ service
		Post-purchase experience

WORKSHEET 6-2

Willingness-to-pay drivers and customer pain points at each stage of the customer journey

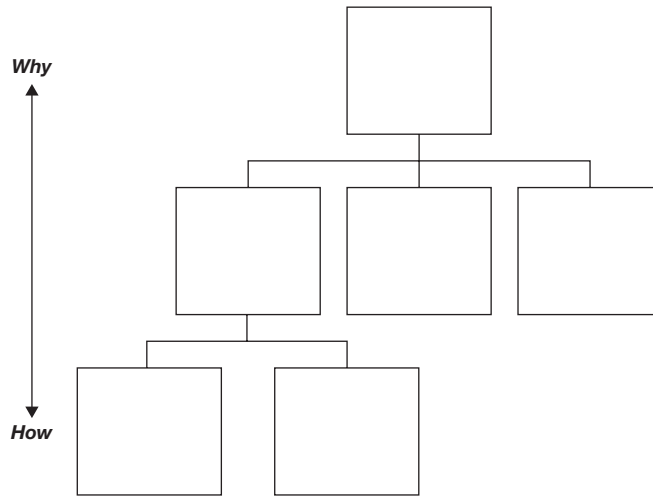
Why does the customer engage in the interaction?	How does the customer go about identifying, ordering, and paying for the desired product?	What products and services are provided to the customer?
Latent need	Awareness of need	Search for options
	Decide on options	Order and pay
	Receive	Experience good/ service
		Post-purchase experience

Information flows at each stage of the customer journey

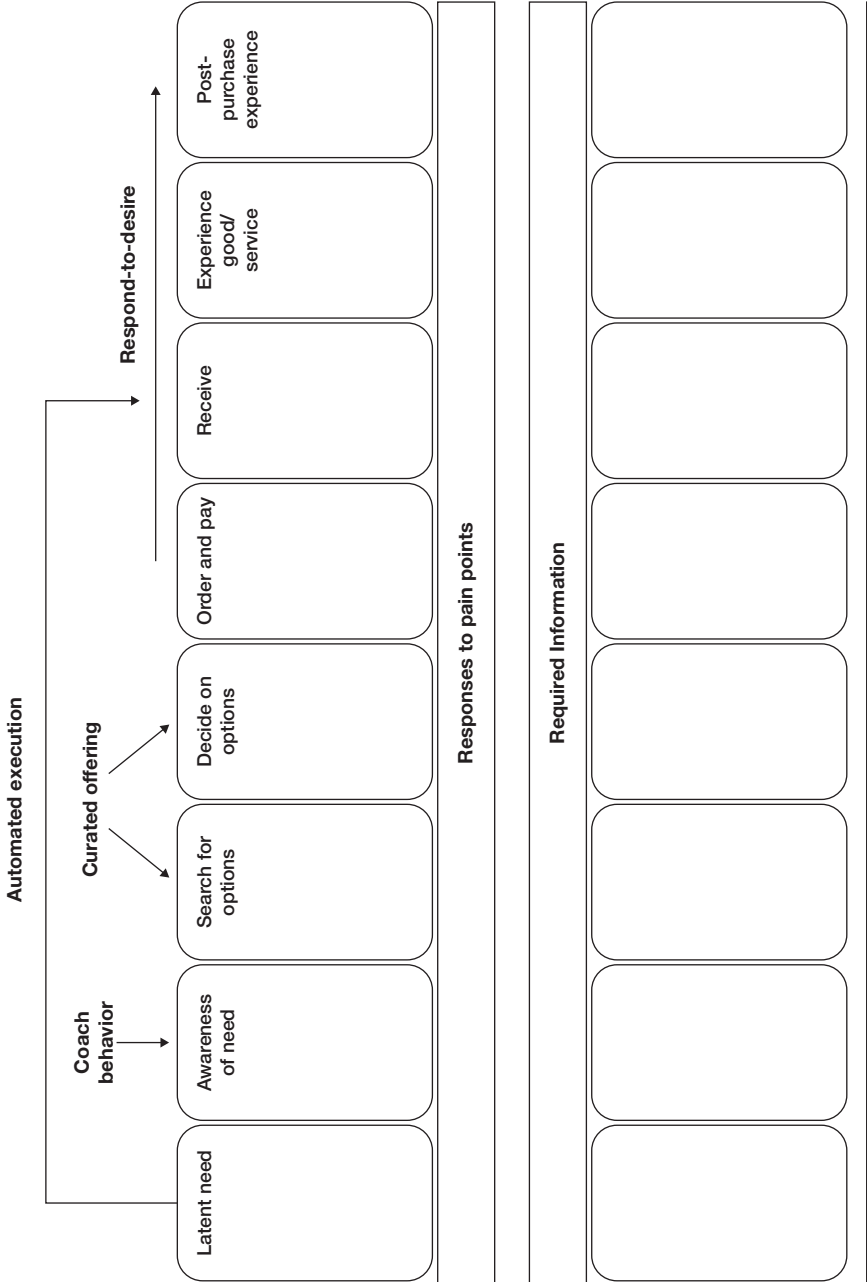
	Why does the customer engage in the interaction?		How does the customer go about identifying, ordering, and paying for the desired product?				What products and services are provided to the customer?		
	Latent need	Awareness of need	Search for options	Decide on options	Order and pay	Receive	Experience good/service	Post-purchase experience	
Description of information									
Trigger									
Frequency									
Richness									
Customer effort									
Action by									
Improvement ideas									

The why-how ladder

In the eyes of the customer, the purpose of the relationship with our firm is to . . .



Responses to pain points and required information



Learning from repeated customer experiences

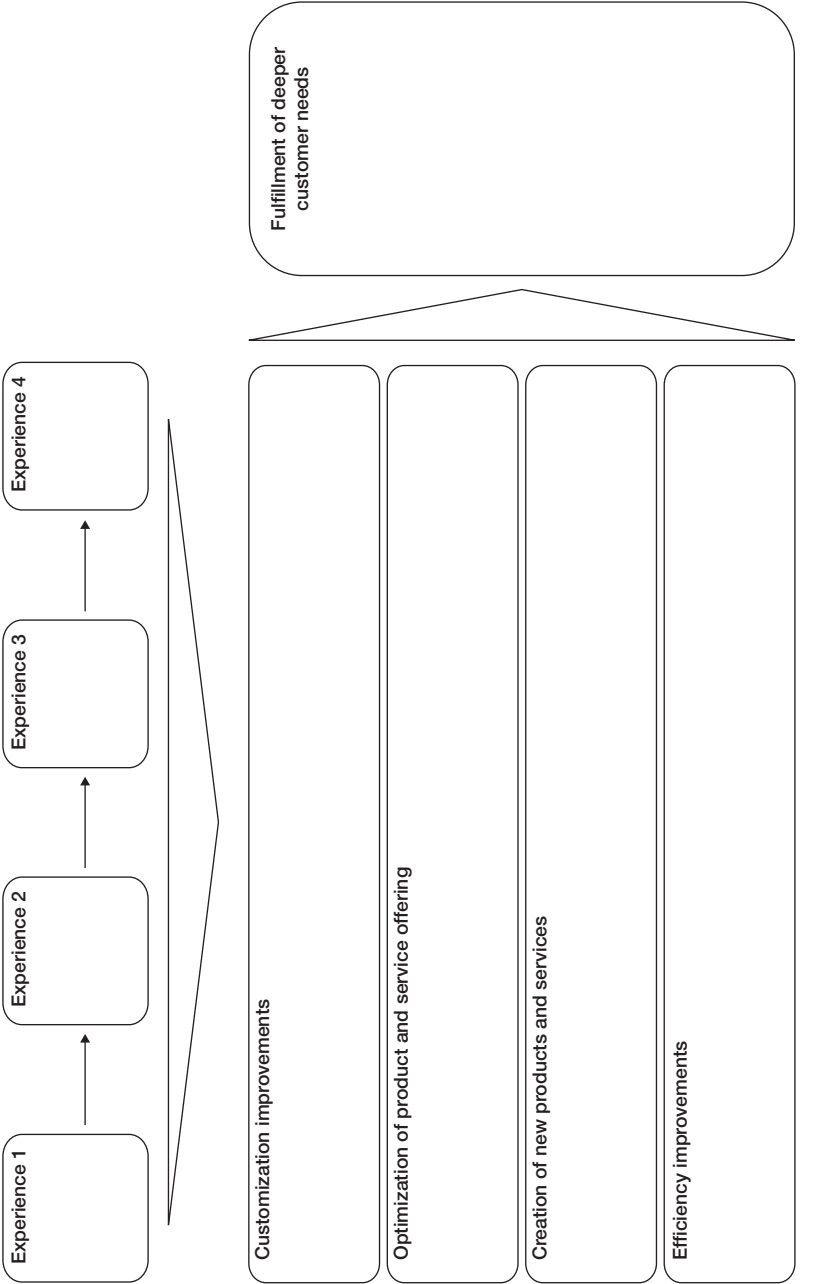


FIGURE 7-1

Connection architecture for a connected producer

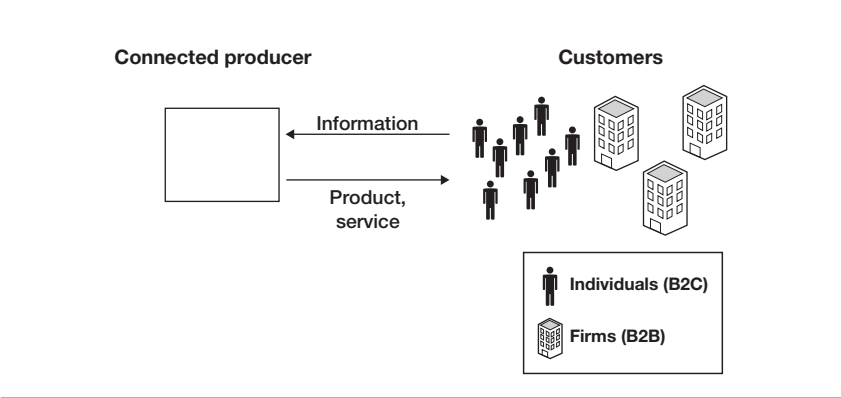


FIGURE 7-2

Connection architecture for a connected retailer

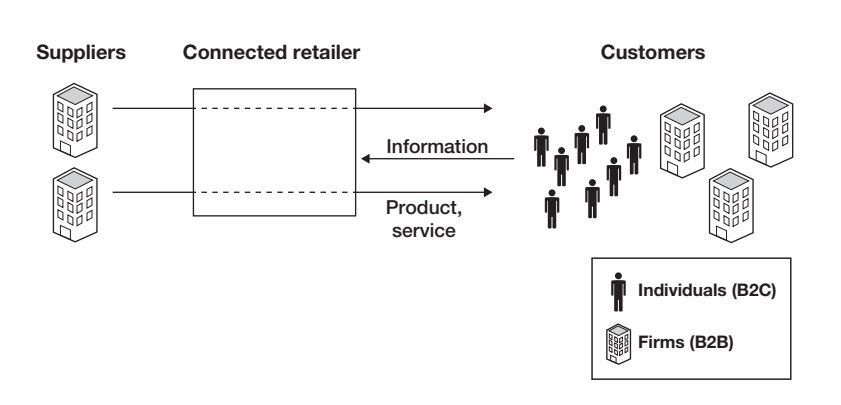


FIGURE 7-3

Connection architecture for a connected market maker

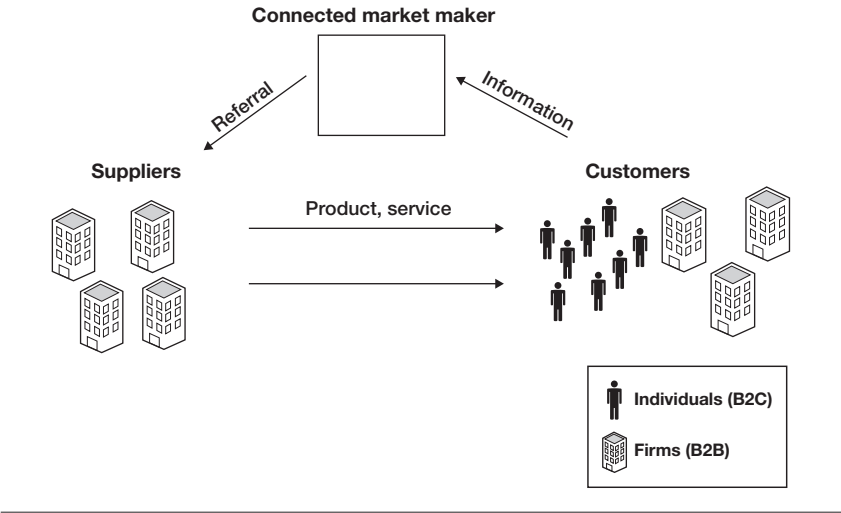


FIGURE 7-4

Connection architecture for a crowd orchestrator

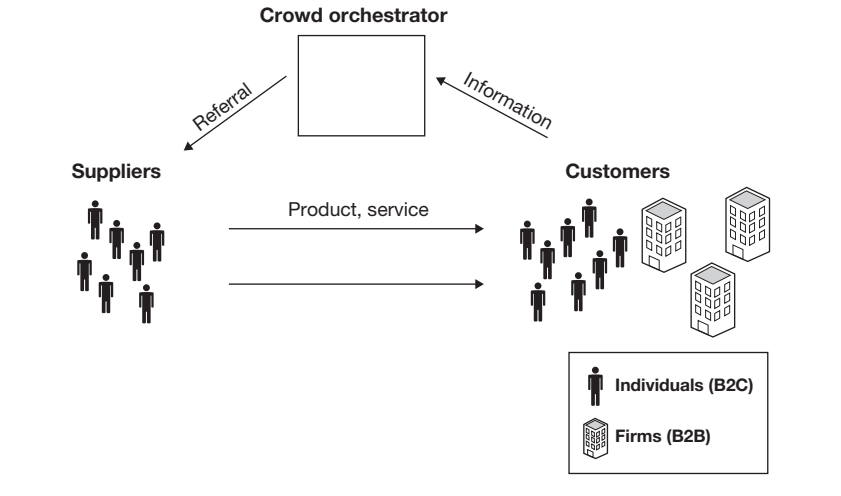


FIGURE 7-5

Connection architecture for a P2P network creator

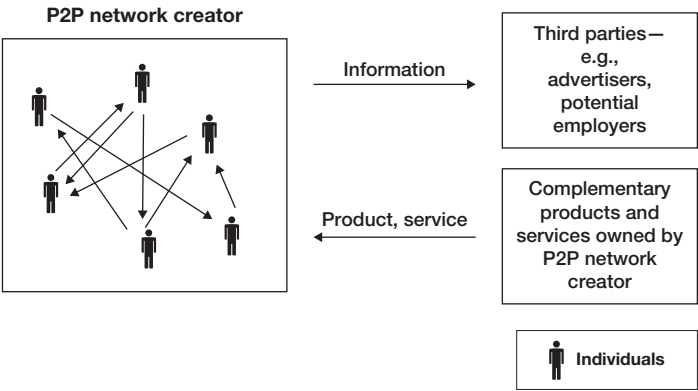


FIGURE 7-6

Connected strategy matrix

	Connected producer	Connected retailer	Connected market maker	Crowd orchestrator	P2P network creator
Respond-to-desire					
Curated offering					
Coach behavior					
Automatic execution					

TABLE 9-1

Two dimensions for deconstructing a connected strategy

	Recognize			Request		Respond		Repeat	Connection architecture	Revenue model
	Become aware of the need	Search and decide on option	Order	Pay	Receive	Experience	After sale	Learn and improve	Connect parties in ecosystem	Monetize customer relationship
Sense	Notice the amount of coffee left in the pantry	Receive information about current coffee prices from nearby retailers	Make sure the desired item is available	Check current cash balance in bank account	Notice arrival of delivery	Notice home owner approaching the front door	Measure change of heart rate and pupil dilation after first sip of coffee	Identify the user whenever she drinks coffee, regardless of location	Sense ordering needs of nearby homes	Check the functionality of the coffeemaker
	Send quantity information to computing system	Get those prices onto the central system	Send order to retailer	Combine account status with projected grocery orders	Send identity information to central system	Send arrival information from door to server	Send information to edge computing system	Send identity and preference information to central system	Pool neighborhood information to one server	Send status report about coffeemaker to service provider

TABLE 9-1 (continued)

Recognize			Request		Respond		Repeat	Connection architecture	Revenue model
	Become aware of the need	Search and decide on option	Order	Pay	Receive	Experience	After sale		
Analyze	Compare with target quantities	Look at prices and volume discounts, potentially factoring in travel plans	If desired item is not available from preferred source, find best alternative	Look for potential account overruns and possible loyalty rewards	Make sure that delivery has been authorized	Identify the person in front of the door	Assess delight of user with coffee brand using physiological measurements	Analyze coffee preferences—e.g., depending on time of day	Make decision when to send coffee-maker replacement
								Evaluate eligibility for group discounts	
React	Decide it is time to start the request module for reordering	Activate order module	Place order for a particular item and provide shipping address	Execute payment	Provide access to pantry or coffee machine	Open the door and activate coffee brewing	Feed results into repeat module	Feed results into request module and help coffee roasters to improve their product	Send coffee machine replacement when necessary
								Negotiate special price with retailer	

Note: Each cell corresponds to a specific subfunction in the coffee-brewing scenario. The table is best read column by column.

FIGURE 9-1

Classification tree for the subfunction “identify a person”

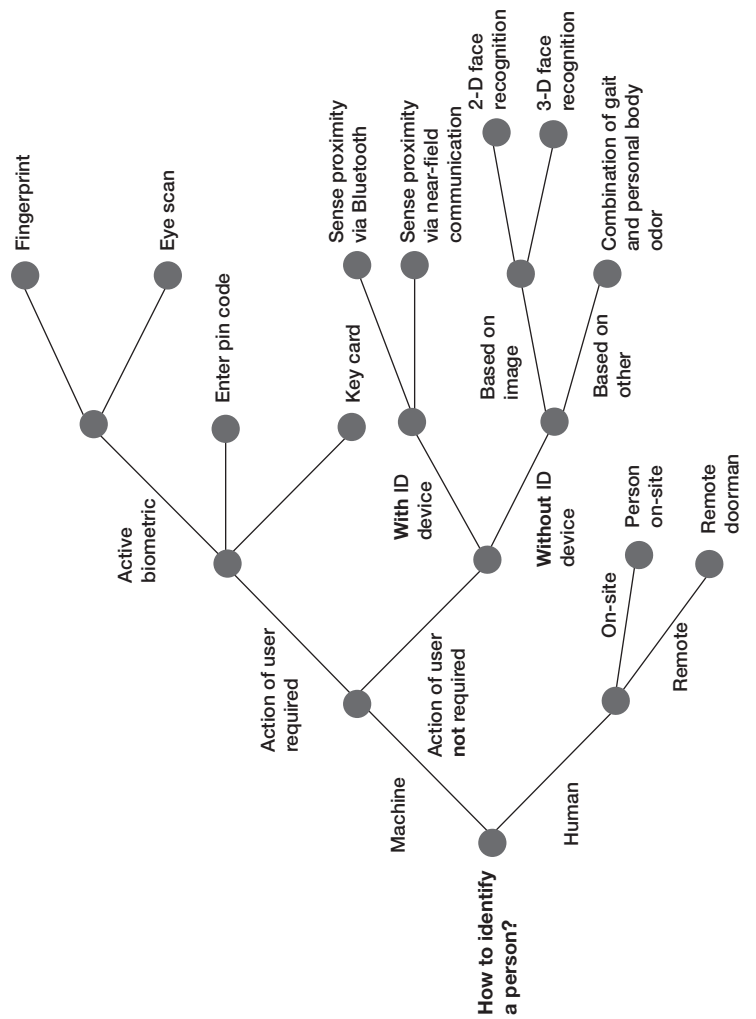


TABLE 9-2

Selection of alternatives for “identify person” subfunction

Technology	Convenience	Safety/ reliability	Cost	Applied in . . .	Comments
Human at door	++	++	--	Hotels	
Human remote	+	+	-	Building access	
Key card	-	++	+	Hospitals	
Enter pin code	-	-	++	Gym	
Fingerprint	-	++	+	Phone	
Eye scan	--	++	-	Global border entry	
Sense device proximity using near-field communication	++	+	+	Car keys	
Sense device proximity using Bluetooth	++	+	+	Wawa	
2-D face recognition	++	+	+	Hotels	
3-D face recognition	++	++	+	High-end phone	
Combination of step pattern and smell	++	?	--	Security dog	Technically not yet feasible

Notes: ++ = very good; + = good; - = poor; -- = very poor; ? = unknown.

The connected strategy matrix: Inventory of your and your competitors' actions

	Connected producer	Connected retailer	Connected market maker	Crowd orchestrator	P2P network creator
Respond-to-desire					
Curated offering					
Coach behavior					
Automatic execution					

The connected strategy matrix: Imagine your company in each cell

	Connected producer	Connected retailer	Connected market maker	Crowd orchestrator	P2P network creator
Respond-to-desire					
Curated offering					
Coach behavior					
Automatic execution					

Deconstruct your connected strategy into subfunctions

	Recognize	Request			Respond			Repeat	Connection architecture	Revenue model
	Become aware of the need	Search and decide on option	Order	Pay	Receive	Experience	After sale	Learn and improve	Connect parties in ecosystem	Monetize customer relationship
Sense										
Transmit										
Analyze										
React										

Possible technological solutions for each subfunction

	Recognize	Request			Respond			Repeat	Connection architecture	Revenue model
	Become aware of the need	Search and decide on option	Order	Pay	Receive	Experience	After sale	Learn and improve	Connect parties in ecosystem	Monetize customer relationship
Sense										
Transmit										
Analyze										
React										

New technological solutions that would enable a new connected strategy

	Recognize	Request			Respond			Repeat	Connection architecture	Revenue model
	Become aware of the need	Search and decide on option	Order	Pay	Receive	Experience	After sale	Learn and improve	Connect parties in ecosystem	Monetize customer relationship
Sense										
Transmit										
Analyze										
React										